MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society

Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3.5: Summary Unsupervised Learning

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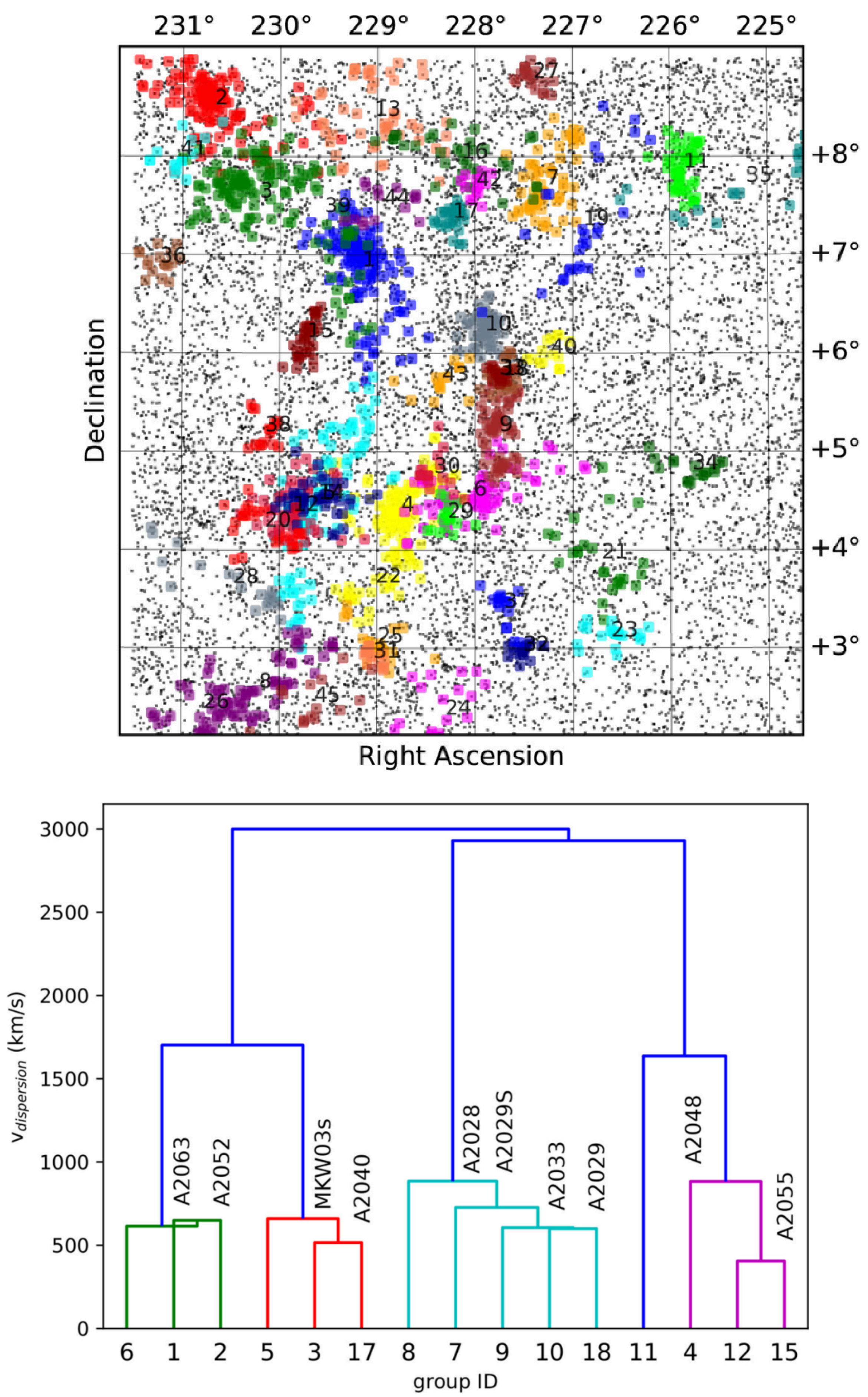
Astronomy: Detection of Galaxy Clusters

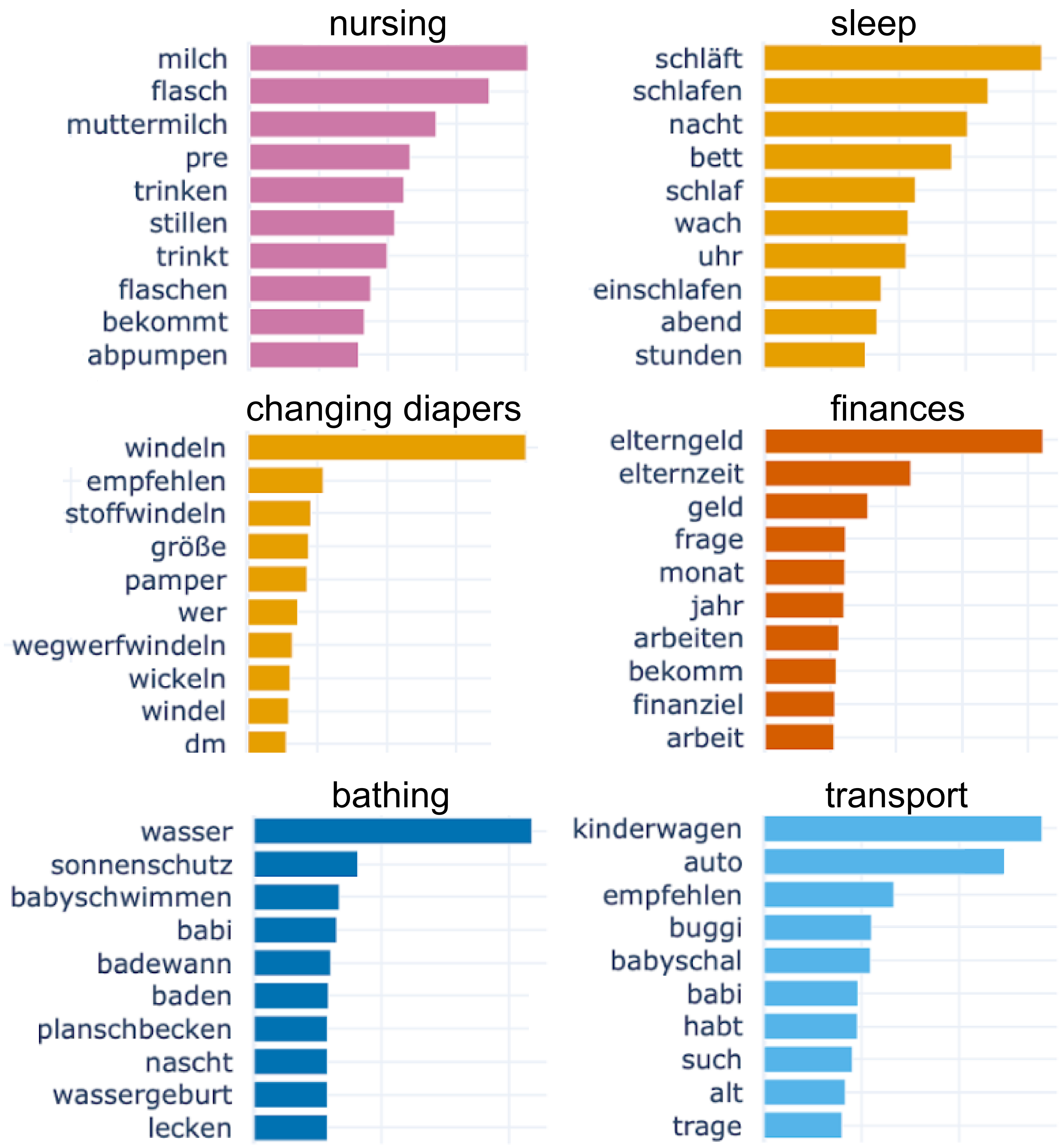
- **Question**: What is the intrinsic structure of galaxy clusters?
- **Approach**:
  - Grouping of galaxies into galaxy clusters in a given sky region using a hierarchical clustering algorithm.
  - Special **distance metric** (binding energy) that describes connections between galaxies.
- **Result**: Hierarchical structure (dendrogram) of the galaxy clusters.



Galaxy clusters detected in the sky region near Supercluster A2029. Source: Yu & Hou, Hierarchical clustering in astronomy, *Astronomy and Computing* (2022).

Sociology: Problems of Young Mothers

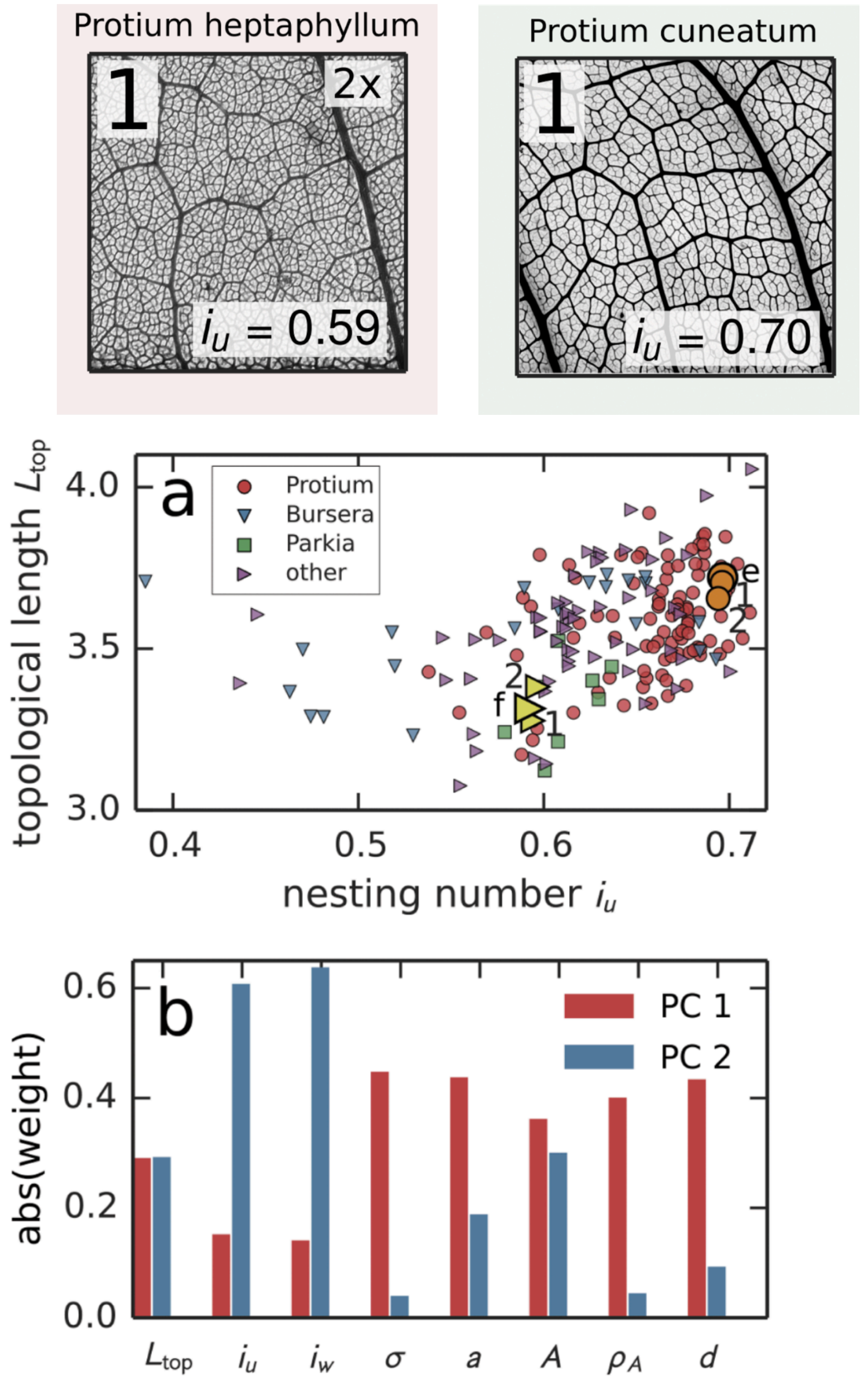
- **Question**: Which topics particularly concern young mothers?
- **Approach**:
  - Identification of relevant public groups on Facebook using keywords and experts.
  - Collection of 840,000 posts.
  - Identification of topics using text categorization (Topic Modeling).
- **Result**: well-separated and interpretable topics.
- **Use**: by the *Association for the Promotion of Child Protection and Child Health in Austria* for the development of a new information brochure.



Topics discussed by young mothers in public German language Facebook groups.
Project work by Magdalena Fritz.

Biology: Characteristics of Plant Leaves

- **Question**: Can we use information from the transport networks of plant leaves to distinguish different species?
- **Approach**:
  - Extraction of transport networks and measurement of network characteristics (topology) and geometric characteristics.
  - Clustering of plant leaves based on characteristics & dimensionality reduction.
- **Results**:
  - The network structure contains information not found in the geometric structure.
  - The PCA clearly shows that geometry and topology cover different dimensions.



Source: Ronellenfitsch et al., Topological Phenotypes Constitute a New Dimension in the Phenotypic Space of Leaf Venation Networks, *PLoS Comp. Biol.* (2015).

Summary Unsupervised learning

- General aim: **discovering unknown patterns in data**.
- Three main applications: cluster analysis, dimensionality reduction, association between observations (covered later) .
- **Cluster analysis** (finding patterns in the observations):
  - **Aim**: find groups of related observations.
  - **Distance and similarity measures** to characterize relatedness.
  - **Algorithms**: hierarchical clustering, K-means, density-based clustering (not covered).
  - **Clustering properties**: exclusiveness, completeness, fuzziness, density- vs. distance-based.
- **Dimensionality reduction** (finding patterns in the features):
  - **Problems with too many features**: storage requirements, curse of dimensionality
  - **Aim**: reduce number of features without losing too much information.
  - **Principal Component Analysis**: recombination of existing features to concentrate variance into only a few dimensions.